Gagan Bhusal
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Education

Bachelor of Engineering in Electronics, Communication and Information **Expected 2027**
Institute of Engineering (IOE), Pulchowk Campus *Nepal*

Experience

Robotics Association of Nepal (RAN) **May 2026 – Present**
Research & Development Intern

- Developing sim-to-real legged-locomotion systems for the Unitree Go2 quadruped and G1 humanoid using Mujoco and Genesis Simulator
- Deployed a trained locomotion policy on the physical Go2, achieving stable and fairly reliable walking gait while resolving observation-space mismatches, two-phase Kp/Kd gain scheduling, and simulation-to-hardware joint mapping
- Deployed motion-imitation policies to the physical G1 using BeyondMimic and GenesisHumanoid, achieving partial sim-to-real transfer; continuing to improve gait quality and stability

Third World Nerd **June 2025 – September 2025**
Machine Learning and Robotics Engineer (Part-Time)

- Co-developed an open-source 8-DOF Quadruped Robot STEM Kit, contributing to hardware integration, servo calibration, and mechanical assembly
- Implemented inverse kinematics for real-time gait generation and real time leg trajectory control in Arduino hardware
- Trained a wake-word detector on a custom voice dataset and integrated it into the robot by deploying it into the ESP-32

Projects

Video Driven Motion Imitation Pipeline for Humanoid Robots **June 2026 – Present**

- University minor project: built a custom Unitree G1 motion-imitation pipeline from scratch using Python, Genesis, Mjlab, and RSL-RL, independent of existing frameworks such as BeyondMimic
- Integrated GVHMR pose estimation, GMR retargeting, and per-skill PPO policies; built a simulation environment with reference state initialization and frame interpolation
- Evaluated policy transfer between Genesis and mjlab, confirming consistent behavior across simulators

Satellite Image Super-Resolution (Research Assistant) | ***Python, PyTorch*** **June 2026 – Present**

- Developing deep learning models to enhance the spatial resolution of satellite imagery using high-resolution UAV imagery as ground truth
- Designed a geospatial preprocessing pipeline with spatially disjoint train, validation, and test splits to prevent geographic data leakage
- Evaluating CNN, GAN, and Transformer based super-resolution architectures to compare reconstruction quality and computational trade offs

Ghost-Net: Edge AI for Acoustic Forest Monitoring | ***PyTorch, TinyML, ESP32*** **March 2026**

- Co-developed a bioacoustic monitoring system using a lightweight CNN and MFCC features to classify 12 environmental sounds, including chainsaws, gunshots, rain, and wind
- Built the data-to-deployment pipeline and achieved 74.6% INT8 accuracy, integrating the quantized model into an ESP32 TensorFlow Lite Micro prototype for on-device inference

English-to-Nepali Machine Translation | *PyTorch* **December 2024 – February 2025**

- Built an encoder–decoder Transformer from scratch in PyTorch for English-to-Nepali translation
- Developed a training pipeline with SentencePiece subword tokenization, learning-rate warmup, and beam-search decoding
- Achieved 22.01 BLEU and 44.62 chrF on 10,812 held-out test sentences using SacreBLEU evaluation

Technical Skills

Languages & Tools: Python, C, C++, Git, Streamlit
ML & Vision: PyTorch, OpenCV, CNNs, GANs, Transformers, Reinforcement Learning, PPO
Robotics: MuJoCo, Genesis, RSL-RL, ROS2, Sim-to-Real Transfer
Edge Deployment: TinyML, TensorFlow Lite Micro, INT8 Quantization, ESP32, Arduino